

Curriculum Vitae

Dr. Arpan Mukherjee



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Personal Information

Date of Birth: 06.04.1994; Nationality: Indian; Gender: Male; Marital Status: Married with 1 kid

Academic Visits & Academic Positions

- Sept 2024 – Present
- MSU-BIT SMBU Joint Research Center of Applied Mathematics, Shenzhen MSU-BIT University, China.
(Moscow State University (MSU), Russia & Beijing Institute of Technology (BIT), China).
 - **Lecturer** (Tenure-Track Assistant Professor)
 - **Hosting Professor:** [Prof. Dr. Hongyu Liu](#), Chair Professor, City University of Hong-Kong, Hong-Kong SAR.
 - **Mentor and Collaborator:** [Univ.-Doz. Dr. Mourad Sini](#), Senior Fellow, RICAM, Austrian Academy of Sciences, Austria.
- Aug 2020 – July 2024
- Johann Radon Institute (RICAM), Austrian Academy of Sciences (ÖAW), Austria.
 - **Post-Doc. Research Scientist** (Dec 2023-July 2024)
Project [FWF-P32660]: Electromagnetism with Extreme Materials: Modeling, Analysis and Applications.
 - **Doc. Research Scientist** (Aug 2020-Nov 2023)
Project [FWF-P32660/36942] : Electromagnetism with Extreme Materials: Modeling, Analysis and Applications / Resolvent Analysis of Sub-wavelength Resonators.
 - **Project Leader:** [Univ.-Doz. Dr. Mourad Sini](#)
- Sep 2019 – Mar 2020
- Department of Mathematics & Steinbuch Center for Computing (SCC), Karlsruhe Institute of Technology (KIT), Germany.
 - **Researcher**

Research Interests

Analysis of Partial Differential Equations; Effective Medium Theories (Homogenization); Mathematical Modeling; Optimal Control and Wave Propagation.

Scientific Works & Publications

Journal Articles

- 1 A. Mukherjee and M. Sini, "Heat Generation Using Lorentzian Nanoparticles: Estimation via Time-Domain Techniques," [SIAM Journal on Multiscale Modeling and Simulation](#), vol. 21, no. 2, pp. 542–597, 2023.
- 2 A. Mukherjee and M. Sini, "Acoustic Cavitation using Resonating Micro-Bubbles. Analysis in the Time-Domain," [SIAM Journal on Mathematical Analysis](#), vol. 55, no. 5, pp. 5575–5616, 2023.

- 3 A. Mukherjee and M. Sini, "Heat Generation Using Lorentzian Nanoparticles. The Full Maxwell System," *SIAM Journal on Applied Mathematics*, vol. 84, no. 1, pp. 285–315, 2024.
- 4 A. Mukherjee and M. Sini, "Time-Dependent Acoustic Waves Generated by Multiple Resonant Bubbles: Application to Acoustic Cavitation," *Journal of Evolution Equations*, vol. 24, no. 90, 2024.
- 5 A. Mukherjee and M. Sini, "Dispersive Effective Model in the Time-Domain for Acoustic Waves Propagating in Bubbly Media," *SIAM Journal on Applied Mathematics*, vol. 85, no. 6, pp. 2508–2542, 2025.
- 6 X. Cao, A. Mukherjee, and M. Sini, "Effective medium theory for heat generation using plasmonics: a parabolic transmission problem driven by the Maxwell system," *Mathematische Annalen*, vol. 393, no. 2-3, pp. 3191–3281, 2025.
- 7 A. Mukherjee and M. Sini, "Dispersive Effective Metasurface Model for Bubbly Media," *Journal of Differential Equations*, vol. 456, no. 113950, 2026.
- 8 A. Mukherjee and M. Sini, "Functional-Analytic Justification of the Time-Domain Foldy-Lax Approximation for Dispersive Acoustic Media: A Feynman-Diagram Viewpoint," 2025 (Submitted). arXiv: [2512.01532 \[math.AP\]](https://arxiv.org/abs/2512.01532).
- 9 A. Mukherjee, S. S. Rodrigues, and M. Sini, "Feedback Stabilization and Tracking for Heat Equations Using Thermo-Plasmonic Nanoparticles as Actuators," 2026 (Submitted). arXiv: [2602.14581 \[math.AP\]](https://arxiv.org/abs/2602.14581).
- 10 A. Mukherjee and M. Sini, "Resonant Microstructures as Dirac-type Actuators for Acoustic Wave Control," 2026 (Submitted). arXiv: [2605.25978 \[math.AP\]](https://arxiv.org/abs/2605.25978).
- 11 A. Mukherjee and M. Sini, "Finite Spectral-Band Optimal Control of Acoustic Waves via Subwavelength Point-Like Resonant Actuators," 2026 (Submitted). arXiv: [2606.24788 \[math.AP\]](https://arxiv.org/abs/2606.24788).

Education

Oct 2021–Nov 2023



Doctor of Philosophy (Ph.D), Johannes Kepler University Linz, Austria.

Industrial Mathematics.

Thesis Title: *Mathematical Analysis of Therapy Modalities using Heat Generation or Acoustic Cavitation.*

Principle Supervisor: [Univ.-Doz. Dr. Mourad Sini](#), Senior Fellow, RICAM, Austrian Academy of Sciences, Austria.

Admin. Co-Supervisor: [Prof. Dr. Haibing Wang](#), School of Mathematics at Southeast University, China.

Marks/ Grades: 1 (**Excellent**)

PhD Courses: Inverse Problems, Mathematical Methods in Electrodynamics, Integral Equation & Boundary Value Problems, Pseudo-differential operators and Fourier integral operators, etc.

Aug 2018 – May 2020



Master of Technology (M.Tech), IIT Madras, India & KIT, Germany.

Industrial Mathematics and Scientific Computing.

Thesis title: *Approximate-Newton Approach for the Intrusive Polynomial Moment Method.*

Supervisors: [Prof. Dr. Sundar S](#) and [Prof. Dr. Martin Frank](#).

Marks/ Grades: 8.91/10

Courses: Mathematical Modelling in Industry, Applied Statistic in R, Numerical PDE, Computer Modelling, Applied Linear Algebra, Numerical Optimization, Data Structure and Algorithm, etc.

Education

- Aug 2015 – May 2017  **Master of Science (M.Sc), Pondicherry University, India.**
Mathematics.
Marks/ Grades: 9.83/10
Courses: Mathematical & Functional Analysis, Measure Theory and Integration, Linear Algebra, Advance Algebra, Advance Topology, Ordinary & Partial Differential Equation, etc.
- July 2011 – July 2015  **Bachelor of Science (B.Sc), Vivekananda Mahavidyalaya, The University of Burdwan, India.**
Mathematics.
Marks/ Grades: 7.2/10
Courses: Differential Equation, Group and Field Theory, Number Theory, 2D and 3D Geometry, Theory of Dynamics, Theory of Metric Space, Linear Programming Problem, etc.

Teaching Experience

- Instructor (Summer 2025)  1. Introduction to Mathematical Theory of Electromagnetism
- Instructor (Winter 2025)  1. Probability Theory & Mathematical Statistics 3. Mathematics for Biological Sciences
2. Linear Algebra (Calculus & Linear Algebra)
- Instructor (Summer 2026)  1. Calculus 2
- Instructor (Winter 2026)  1. Probability Theory & Mathematical Statistics 3. Linear Algebra
2. Engineering Mathematical Analysis I
(Artificial Intelligence students)

Student Supervision

- Under-graduate  1. Mei Jia, Introduction to Scattering Theory: Uniqueness Results and Expansion Theorems for Solutions of the Helmholtz Equation.
2. Du Xin, Mathematical Modeling and Analysis of Metamaterials.

Miscellaneous Academic Success

Fellowships & Awards

- 2026  Selected as Distinguished Academic Scholar (Specially Appointed Position (C Category)) in Longgang District of Shenzhen MSU-BIT University, China.
- 2020  Awarded Doctoral Fellowship based on the project FWF-P32660: “**Electromagnetism with Extreme Materials: Modeling, Analysis and Applications**”, Austrian Science Fund (FWF), Austria.
- 2019-2020  Awarded a Scholarship "Combined Study and Practice Stays for Engineers from Developing Countries (KOSPIE) with Indian IITs", Deutscher Akademischer Austauschdienst - **DAAD**, Germany.
- 2018-2019  Awarded a **Fellowship Half Time Teaching Assistant (HTTA)**, IIT Madras & Ministry of Human Resource Development (MHRD), Chennai, India.

Miscellaneous Academic Success

- 2018  Awarded **DST-INSPIRE Fellowship (Junior Research Fellowship)**, Govt. of India, Ministry of Science and Technology, Dept. of Science and Technology (DST), India. (Not availed)
- 2016-2017  Awarded a **Merit Scholarship for obtaining good academic performance in MSc**, Pondicherry University (A Central University), Pondicherry, India.
- 2011-2016  Awarded a **Scholarship for Higher Education – Innovation in Science Pursuit for Inspired Research (INSPIRE)**, Govt. of India, Ministry of Science and Technology, Dept. of Science and Technology (DST), India.

Certification

- 2018  Qualified in **Graduate Aptitude Test in Engineering (GATE)**, IITs and IISc, All India Rank (AIR) 469, GATE Score 464.
- 2015  Qualified in **Joint Admission Test (JAM)**, IITs and IISc, India.

Research Activity

Presentations (Invited, conference or Seminar)

- 2025  The 3rd HKSIAM Biennial Conference, July 7-11, 2025, The Chinese University of Hong Kong and Hong Kong Society for Industry and Applied Mathematics (HKSIAM), Hong-Kong SAR.
-  The 1st Sino-Belarusian Mathematics and Applied Mathematics Conference, April 3, 2025, Shenzhen MSU-BIT University, China.
-  The 2nd SMBU Sino-Russian Conference on Mathematics, March 22, 2025, Shenzhen MSU-BIT University, China.
- 2024  14th AIMS Conference, December 16 – 20, 2024, New York University Abu Dhabi and AIMS, Abu Dhabi, UAE.
-  CityU-SJTU Joint Workshop on Computational Mathematics, November 23 – 24, 2024, City University of Hong-Kong, Hong-Kong SAR.
-  9th European Congress of Mathematics (ECM 2024), July 15 - 19, 2024, European Mathematical Society, Spain.
-  8th International Conference on Applied Mathematics (ICAM 2024), May 28- June 1, 2024, City University of Hong-Kong, Hong-Kong SAR.
-  Department of Mathematics and Statistics Seminar, "Mathematical Analysis of Therapy Modalities using Acoustic Cavitation", January 18, 2024, IIT Tirupati, India.
- 2023  17th International Workshop on Optimization and Inverse Problems in Electromagnetism (OIPE 2023), September, 17 – 20, 2023. TU Graz, Austria.
-  11th Applied Inverse Problems Conference 2023 (AIP 2023), September 4-8, 2023, University of Göttingen, Germany.
-  3rd Alps-Adriatic Inverse Problems Workshop 2023 (AAIP 2023), July 5-7, 2023, Alpen Adria Universitaet Klagenfurt, Austria.
- 2022  International Conference on Analysis, Inverse Problems, and Applications, July 18-21, 2022, IIT Madras, India. (Paper accepted for Presentation).
- 2019  "Stability of miscible displacements in Porous Media: Rectilinear flow", 2019, IIT Madras, India.
- 2018  "Comparative Study of Various Image Noise Reduction Techniques and A Small Survey on PSNR and MSE", 2018, IIT Madras, India.

Research Activity

- "A Small Survey about Traffic Flow Problem", 2018, IIT Madras, India.

Conference/Workshop Organizing

- 2025 ■ Co-organizer, Minisymposium: "Inverse Problems for Evolution Equations", 12th Applied Inverse Problem Conference (AIP2025), Rio de Janeiro, Brazil, July 28-August 1, 2025.

Conference Participation

- 2022 ■ Special Semester on Tomography Across the Scales, Oct 3 - Dec 2, 2022, RICAM, Austrian Academy of Sciences, Austria.
 - Austrian Numerical Analysis Day, May 4-6, 2022, RICAM, Austrian Academy of Sciences and JKU Linz, Austria.
- 2021 ■ Special Semester on Tomography Across the Scales, October 11-15, 2021, RICAM, Austrian Academy of Sciences, Austria.
- 2013 ■ MTTS (Mathematical Training and Talent Search) Program, IIT Bhubaneswar, School of Basic Sciences, funded by NBHM (National Board of Higher Mathematics), India.

Skills

- Languages ■ Strong reading, writing and speaking competencies for English, Hindi (National), Bengali (Native).
- Coding ■ Python, Julia, Matlab, R
- Misc. ■ LATEX typesetting and publishing.

References

Univ.-Doz. Dr. Mourad Sini

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Prof. Dr. Hongyu Liu

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City University of Hong Kong, Kowloon,
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Prof. Dr. S. Sundar (Director, NIT Mizoram)

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